

Aryan Sachdev

EDUCATION

University of Illinois at Urbana-Champaign
B.S. in Molecular and Cellular Biology

May 2028 (Expected)
Urbana-Champaign, IL

WORK EXPERIENCE

K.V. Prasanth Lab — University of Illinois
Research Assistant

Oct 2025 – Present
Urbana-Champaign, IL

- Built an RNA-seq pipeline across 5 hypoxia-versus-normoxia breast cancer models to identify hypoxia-induced RNA-binding proteins and lncRNAs as CRISPRi screen candidates.
- Corrected ~60 misdirected guide designs by ranking candidates at the isoform and promoter level before the library went to the National Cancer Institute.
- Validated top candidates by RT-qPCR and established a stable MCF7 dCas9-ZIM3 CRISPRi cell line for an upcoming hypoxia growth-based functional screen.

The EYE Center (Dr. Samir Sayegh)
Clinical Technician Volunteer

Aug 2026 – Present
Champaign, IL

- Run patient pre-exams before physician visits: histories, OCT and retinal imaging, refraction and keratometry, and IOLMaster and ultrasound biometry.

IronStreet Advisors
Consultant

Sep 2026 – Present
Remote

- Build web and social media presence for a healthtech advisory firm serving digital health, MedTech, and AI-enabled care companies.

Center for AI Innovation (NCSA)
AI Development Intern (Students Pushing Innovation program)

May 2025 – Oct 2025
Urbana-Champaign, IL

- Built a Canvas LMS-integrated course assistant for Illinois Chat, the campus-wide AI assistant, which won First Place in the Innovative Tool category at Instructure's Academic Excellence Awards.
- Engineered a React and Node.js application with real-time streaming and Canvas LTI integration; optimized retrieval so answers come from the right course materials.

Cannabis Research Institute — University of Illinois Chicago
Research Assistant

Jun 2025 – Aug 2025
Chicago, IL

- Detected Hop Latent Viroid (HLVd), an RNA pathogen affecting crop yield, using RNA extraction and precipitation, RT-qPCR, and genotyping.
- Modeled how single amino acid changes in terpene synthase enzymes affect function using AlphaFold, Rosetta, and PyMol.

PUBLICATIONS

- Om A, Boric A, **Sachdev A**, et al. "Clinical Management of Radiation Dermatitis in Breast Cancer Patients: A Systematic Review and Meta-Analysis of Randomized Controlled Trials." *In revision*.

HONORS, AWARDS & INVOLVEMENT

- **QCB Villa Cisse Scholar (NSF)**: One of five selected for a 400-hour independent summer research program.
- **IEEE Undergraduate Oral Presenter**: One of four selected to present at IEEE ETHICS-2025, Northwestern University.
- **Dean's List**: University of Illinois at Urbana-Champaign, Spring 2026.
- **Captain, UIUC Club Tennis** (130 members): Sectional Champion; 5th at national championships.

PROJECTS

- **Athen.AI** (Co-Founder, 2025–2026): Platform built with physicians for safe clinical AI adoption; its PHI de-identification pipeline covers all 18 HIPAA Safe Harbor identifiers and blocks upload on failure.
- **ClearAF** (Co-Founder, 2025–Present): Telehealth platform built with a practicing dermatologist, pairing AI skin analysis with treatment history for personalized Rx plans; retrained on more varied skin tones after weaker results on darker skin.

SKILLS

- **Laboratory Techniques**: RNA extraction, RT-qPCR, RNA-seq, cell culture, genotyping, gel electrophoresis.
- **Programming & AI**: Python, R, Java, Node.js, React/Next.js, FastAPI, PostgreSQL; LLM development and fine-tuning.